

07 — Certification, Error Bounds, and Ranking Stability

SEION-KGR v26 MAX

Campaign ID: gate12-closeout-2026-08-01

Canonical commit: 6af3c35271ae2ffab41ecba2aad098d1988fdc0c

Branch: campaign/gate12-closeout

Date: 2026-08-01

Purpose: state exactly what is certified, under what checked assumptions, and report coverage honestly separate from accuracy.

Contents

1	The four namespaces	1
2	What is certified (narrow, but real)	1
3	What is honestly NOT certified	1
4	Coverage, reported separately from accuracy	2
5	Consequence for any future compressed-model deployment	2

1 The four namespaces

`certified_bound` (valid under checked assumptions), `empirical_majorant` (measured, never proved), `empirical_error_predictor` (fitted/measured proxy, e.g. the closure-leakage sample mean), `observed_error` (direct measured discrepancy, e.g. ranking margin). These are structurally separate Python objects in `seion_kgr/certification.py` — never merged in code, so they cannot be merged by accident in a report either.

2 What is certified (narrow, but real)

Theorem (elementary, PROVED). If $\|\mathbf{s} - \tilde{\mathbf{s}}\|_\infty \leq \varepsilon$ and $s_i - s_j > 2\varepsilon$, then $\tilde{s}_i > \tilde{s}_j$. Direct triangle-inequality argument.

This campaign’s contribution is the *chain* that produces a checked ε for one real (if narrow) case: when the score path involves no rank-reducing projector and no nonlinear envelope, the deployed computation is identical to the ambient computation — nothing was approximated — so $B_{\text{state}} = 0$ exactly, and $\varepsilon = 0$. Any strictly positive margin then certifies the ranking. This holds for base-expert-only scoring, and (via duck typing on the shared A, B, C, O weight structure) for the seionic scalar branch used alone.

The exact operator-norm bound used when a CP-shaped law IS present in the certified path:

$$M_{\text{bound}} = \|O\|_{\text{op}} \|A\|_{\text{op}} \|B\|_{\text{op}} \|C\|_{\text{op}}$$

(top singular value of each factor, exact SVD, not sampled), a valid upper bound via the elementwise-product Cauchy–Schwarz inequality ($\|u \odot v \odot w\|_2 \leq \|u\|_2 \|v\|_2 \|w\|_2$, since $\|\cdot\|_\infty \leq \|\cdot\|_2$) composed with each factor’s operator norm. Verified never exceeded across 500 random unit-norm probes (`tests/kgr/test_certification.py`).

3 What is honestly NOT certified

The moment a rank-reducing `StiefelProjector` (`rank < dim`) or the path reasoner’s `Layer-Norm+tanh` envelope enters the score path, certification returns `NOT_CERTIFIED` unconditionally — verified with a test using a margin of 2000 that still refuses certification, proving margin size cannot compensate for a missing assumption check. The two specific missing assumptions:

- **Closure leakage operator norm** $\rho_\mu = \|(I - P)\mu(P \cdot)\|_{\text{op}}$ — no checked bound exists anywhere in this repository (assumption A4). The sample-mean closure-leakage ratio used in training (`docs/definitions/projectors.md`) is explicitly a proxy, never an operator-norm certificate.
- **Nonlinear-envelope Lipschitz constant** for `LayerNorm(tanh(...))` — no checked bound exists (assumption A6).

4 Coverage, reported separately from accuracy

`certify_evaluation_split` is a structurally separate pass from `evaluate.evaluate()` — coverage is never blended with MRR/Hits@K in the same call, satisfying the mandate's explicit "report coverage separately from accuracy." Two integration tests (`tests/kgr/test_certification.py`) demonstrate the honest coverage swing on identical underlying data: a base-expert-only `SeionKGRv26` instance shows 100% `certified_rank_stable_coverage` (every non-tied query certifies); the identical setup with the path reasoner enabled shows exactly 0% — not a bug, the correct behavior of a chain that refuses to certify what it has not checked.

No coverage study was run on real benchmark data in this campaign. The two integration tests above use the 6-entity synthetic `tiny_kg` fixture, not FB15K-237/WN18RR. This is stated explicitly so the 100%/0% figures are not misread as a benchmark-scale result.

5 Consequence for any future compressed-model deployment

Until A4 or A6 is resolved (a real operator-norm bound for closure leakage, or a real Lipschitz bound for the nonlinear envelope), *no* certified compression claim is available for any SEION-KGR configuration that uses the path reasoner or a rank-reduced projector. This is the honest state of the art in this repository as of this campaign; closing it is future work, listed in `09_negative_results_limitations_and_open_problems.pdf`.